Artifact 20 — AI Unit Economics

Strategy · Stage 5 · Business Model Validation

DOCUMENT DETAILS

Project	Home-Care-AI	**Stage**	Strategy → Stage 5 (Business Model Validation)
Date	2026-03-27	**Methodology**	Agentic Action cost decomposition; Gross Margin projection; Token Efficiency Strategy; Scalability Threshold; Infrastructure Scaling Roadmap
Input	Artifact 12 (Startup Canvas — pricing tiers, cost preview, Claude Haiku/Sonnet split), Artifact 17 (Remediation — template-based v1, no LLM external content, Channel Wrapper), Artifact 18 (Partnership Mapping — PAC estimates, SMS gateway), Artifact 19 (Positioning — cost advantage 3/5 moat, Artifact 20 pending note), CLAUDE.md Article V HS-STRAT-03 (token budget → model selection)		
Regulatory	Australian Privacy Act 1988 (APP). HIPAA-grade security applied as design floor. All infrastructure in ap-southeast-2 (Sydney) — no cross-border cost or compliance premium.		
Feeds into	`agentic-logic-spec` (HS-STRAT-03 — model selection and token budget per agentic action); `create-prd` (cost constraints, scalability thresholds)		

Skill Pre-Condition Check (CLAUDE.md Article IV Rule 6 — "Price Last"): - ■ Compliance costs confirmed: Artifact 16 (DPIA scope), Artifact 17 (SMS gateway selected, LLM-external content deferred) - ■ Partnership complexity confirmed: Artifact 18 (PAC estimates, SMS provider, Google Maps APP 8 pending) - ■ Feature set confirmed: Artifact 19 (v1 scope = rule-based matching + template notifications + SPP data model) - ■ Pricing confirmed: Artifact 12 §11 ($199 Solo / $149 Team / $119 Agency per coordinator seat/month AUD)

All three prerequisites are satisfied. Unit economics is valid to run.

Critical v1 architecture note (from Artifact 17 CRIT-02 resolution): v1 uses template-based string interpolation for all external-facing content. No LLM generates text transmitted to carers, clients, or families. The Smart Match Engine is rule-based (Artifact 12 §8: "no ML required in v1 — rule-based scoring is sufficient and auditable"). This means LLM inference costs in v1 are near-zero for the critical delivery path — a direct consequence of the compliance-first architecture decision. This is the most important finding in this artifact: the CRIT-02 resolution is a gross margin windfall.

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§01 — Unit of Analysis: The Agentic Action

The atomic unit of AI work in Home-Care-AI v1 is the **Vacancy Incident Resolution** — the full automated workflow from vacancy detection to coordinator approval to notification dispatch to audit log closure.

Action inventory per vacancy incident

Step	Action	Level	LLM?	Primary Cost Driver
1	Vacancy event detection (carer sick, roster gap)	**L1 Informer**	**No — event trigger**	Negligible compute
2	Qualification gate (hard filter: carer certifications vs. client requirements)	**L1 Informer**	**No — rule-based DB query**	DynamoDB read
3	Proximity score (Google Maps Distance Matrix)	**L1 Informer**	**No — API call**	Google Maps API
4	SPP match score (familiarity + preference weighted algorithm)	**L1 Informer**	**No — deterministic algorithm**	DynamoDB read
5	Shortlist generation and ranking (top 3 candidates)	**L1 Informer**	**No — sort algorithm**	Negligible compute
6	HITL coordinator presentation (3-Tap Flow push notification)	**L2 Verifier**	**Haiku — agentic orchestration layer**	Haiku inference + push
7	Coordinator approval (ACT-A-02 — taps "Approve")	**L2 Verifier**	**Haiku — state machine update**	Haiku inference
8	Notification dispatch (SMS carer → client → family, E-3 gate enforced)	**L1 Informer**	**No — template interpolation**	**SMS gateway** (dominant cost)
9	Carer briefing assembly (structured SPP field assembly, P-3 branch)	**L1 Informer**	**No — template, CRIT-02**	SMS gateway
10	Audit log entries (8–12 entries per incident, CloudTrail)	**L1 Informer**	**No — structured write**	CloudTrail ingestion
11	VACANCY_UNRESOLVED escalation (if no candidate accepted)	**L3 Escalator**	**Sonnet — complex orchestration**	Sonnet inference

Where LLM appears in v1

- Steps 6–7: Haiku orchestration layer manages state transitions, routes decisions, formats approval card
- Step 11 (rare): Sonnet handles L3 escalation reasoning (VACANCY_UNRESOLVED — complex, low-frequency)
- LLM does NOT appear in: matching logic, notification content, briefing content, audit logging

This is the consequence of the CRIT-02 resolution and the rule-based v1 match engine. The LLM is the orchestration wrapper, not the intelligence engine.

§02 — Inference Cost (Token Spend)

2.1 Hidden System Prompt Overhead

Per CLAUDE.md Article V (mandatory calculation):

Component	Token Estimate	Notes
APP compliance constraints	250–350	APP 8 cross-border rules, PHI handling, minimum necessary instruction. Leaner than HIPAA equivalent (scheduling context, not clinical care)
Persona lock + role definition	100–150	"Care coordination scheduling agent" — narrow scope, fewer edge cases than clinical agent
Agentic safety level definitions (L1/L2/L3)	150–250	HITL trigger conditions, VACANCY_UNRESOLVED protocol, E-3 gate enforcement
Few-shot examples (scheduling decisions)	300–500	Required for consistent shortlist presentation format and coordinator approval flow
SPP context injection (dynamic per session)	400–800	Candidate SPP match data, client preference summary. Varies with history depth.
Output format constraints (structured JSON)	80–120	Action decision + state update format
Total system prompt floor	**1,280–2,170 tokens**	Well within CLAUDE.md 1,550–4,000 floor; using **1,550 token floor** per Article V mandatory calculation

System Prompt Overhead Cost (Claude Haiku, 1,550 token floor)

```
Without caching: 1,550 × $0.80/1M = $0.00124 per call
With prompt caching (80% hit rate, 10% of full price for cache):
  → Cache hit (80%): 1,550 × $0.08/1M = $0.000124
  → Full price (20%): 1,550 × $0.80/1M = $0.00124
  → Weighted: (0.80 × $0.000124) + (0.20 × $0.00124) = $0.000099 + $0.000248 = $0.000347/call
```

Prompt caching reduces system prompt overhead by **72%** — strongly recommended as a Day 1 implementation.

2.2 Per-Action Token Breakdown

Action Type A — L1/L2 Routine (Haiku): Vacancy Orchestration

Frequency: 2–3 Haiku calls per vacancy incident (state trigger, coordinator presentation, approval processing)

Token Component	Tokens	Cost (Haiku, with caching)
System prompt (cached, 80% hit)	1,550	$0.000347
Dynamic context (vacancy data + SPP + shortlist)	450	450 × $0.80/1M = $0.00036
Output (action decision + JSON)	200	200 × $4.00/1M = $0.0008
Total per Haiku call	**2,200 tokens**	**$0.0015**
Total for 3 Haiku calls per incident	**6,600 tokens**	**$0.0045**

Action Type B — L3 Escalation (Sonnet): Vacancy Unresolved

Frequency: 2–3 times/month per agency (estimated ~20% of incidents where top 3 candidates all decline or are unavailable)

Token Component	Tokens	Cost (Sonnet)
System prompt (not cacheable — low frequency)	1,550	1,550 × $3.00/1M = $0.00465
Escalation context (all candidates tried, reasons, coordinator context)	1,200	1,200 × $3.00/1M = $0.0036
Output (escalation reasoning + VACANCY_UNRESOLVED decision)	400	400 × $15.00/1M = $0.006
Total per Sonnet call	**3,150 tokens**	**$0.01425**
Total for 2.5 Sonnet calls per month per agency	**7,875 tokens**	**$0.036**

Action Type C — SPP Completeness Analysis (Haiku): Daily Check

Frequency: 1 call/day per agency = 30 calls/month

Token Component	Tokens	Cost (Haiku, with caching)
System prompt (cached)	1,550	$0.000347
SPP state data	350	350 × $0.80/1M = $0.00028
Output (completeness flags for coordinator)	150	150 × $4.00/1M = $0.0006
Total per call	**2,050 tokens**	**$0.00123**
Total for 30 calls per month per agency	**61,500 tokens**	**$0.037**

2.3 Total Monthly Inference per Agency

Action Type	Calls/Month	Cost/Call	Monthly Inference Cost
L1/L2 Haiku (vacancy orchestration)	36 (12 incidents × 3 calls)	$0.0015	$0.054
L3 Sonnet (vacancy unresolved)	2.5	$0.01425	$0.036
Haiku (SPP completeness)	30	$0.00123	$0.037
Total monthly inference	**68.5**	—	**$0.127**

Finding: AI inference costs $0.13/agency/month in v1. This is not the cost problem. This is the compliance-first architecture dividend — template-based external content and rule-based matching eliminate the expensive LLM calls from the critical path.

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§03 — Storage Cost (APP-Compliant Australian Infrastructure)

All storage in **AWS ap-southeast-2 (Sydney)**. No cross-border transfer. No APP 8 premium on storage (data stays domestic).

Component	Per Agency/Month	Notes
DynamoDB (SPP + carer profiles + availability index)	$1.20	~120 KB active data per agency at beachhead scale. On-demand billing.
S3 write-once audit log (7-year retention)	$0.45	~8 entries × 12 incidents × 0.5 KB = 48 KB/month. At $0.023/GB/month.
CloudTrail ingestion	$0.02	48 KB/month × $0.50/GB ingested
KMS key management	$1.00	1 customer-managed key per agency
CloudWatch monitoring + alerting	$0.80	Log ingestion + metric alarms
Lambda compute (state machine functions)	$0.12	~500 invocations/month (incidents + SPP + audit). Well within free tier at beachhead.
API Gateway (coordinator app calls)	$0.08	~200 API calls/month per coordinator
S3 object storage (SPP documents, carer certifications)	$0.05	~2 GB at $0.023/GB
Total monthly storage + infrastructure per agency	**$3.72**	

Encryption overhead (+10–15% compute): +$0.37 → **rounded total: $4.10/agency/month**

§04 — Channel / Notification Cost

Component	Per Incident	Incidents/Month	Monthly Cost/Agency
SMS to carer (AU domestic, MessageMedia / AWS SNS)	$0.08 AUD	12	$0.96
SMS to client (APP 8 compliant, AU domestic)	$0.08 AUD	12	$0.96
SMS to family (AU domestic)	$0.08 AUD	12	$0.96
SMS: VACANCY_UNRESOLVED coordinator alert	$0.08 AUD	2.5	$0.20
Google Maps Distance Matrix API (1 call × 5 candidates)	$0.025 AUD	12	$0.30
Push notification to coordinator (AWS SNS mobile push)	$0.001 AUD	12	$0.012
Total channel / notification cost	**$0.27/incident**		**$3.42/agency/month**

<i>Note: SMS is the single largest variable cost driver — 83% of per-incident costs. AI inference ($0.004/incident) is 1.5% of per-incident cost. This is the key ratio: ***SMS costs 60× more than AI inference per incident.*** *</i>*

§05 — Cost-per-Agency Summary (Variable COGS)

Cost Component	Monthly Cost/Agency	% of Variable COGS
AI inference (Haiku + Sonnet)	$0.127	1.7%
Storage + infrastructure (AWS)	$4.10	55.4%
Channel costs (SMS + Maps + push)	$3.42	**46.2%** — rounding to total
API overhead + monitoring	$0.08	1.1%
Total variable COGS per agency/month	**$7.41 AUD**	100%

Consistent with Artifact 12 §10 preview estimate of $5–15/agency/month. The range reflects incident volume variation: low-volume agency (6 incidents/month) ≈ $5.20; high-volume agency (25 incidents/month) ≈ $14.80.

The dominant cost insight: Variable AI infrastructure is not the margin problem. At $0.13/month/agency for inference, token costs are negligible at any realistic beachhead scale. The real margin levers are SMS unit price negotiation and compliance fixed cost amortization (§06 below).

§06 — Compliance Infrastructure Cost (Fixed Period Cost)

Compliance infrastructure is a fixed operating cost, not a per-unit variable cost. It does not scale with agency count until v2+ feature expansion. Per Artifact 16 and Artifact 12:

Cost Item	Estimate (AUD/month)	Classification	Notes
Privacy counsel retainer (ongoing DPIA, APP guidance)	$2,000–$4,000	Fixed operating	Reduces after DPIA-01–07 complete. Ongoing for new features.
DPIA completion amortization ($15K one-time / 24 months)	$625	Amortized	DPIA-01 through DPIA-07 from Artifact 16
E-1 anti-discrimination legal opinion amortization	$167–$208	Amortized	$3K–$5K one-time, 24-month amortization
APP 8 counsel (SC-07 Google Maps review, v2 WhatsApp unlock)	$83–$167	Amortized	$2K–$4K one-time
CRIT-04 AX-01 engineering confirmation + schema audit	$200–$400	One-time (before Sprint 1)	Not recurring
Total fixed compliance per month	**$3,075–$5,000**	Fixed	Peak at launch; reduces as DPIAs are completed

Compliance cost per agency at different scale points

Agencies	Compliance Fixed Cost	Per-Agency Compliance	% of $199 Revenue
5	$4,000 (midpoint)	$800	**402% — existential**
10	$4,000	$400	**201% — deeply underwater**
25	$4,000	$160	**80% — survivable with team seat revenue**
50	$3,500 (DPIAs completing)	$70	**35%**
100	$3,200	$32	**16%**
200	$3,000 (steady state)	$15	**7.5%**

Insight: Compliance is a start-up tax that amortizes sharply after 25 agencies. Below 25 agencies, compliance cost per agency exceeds the entire subscription price. The founding team must treat compliance as fixed overhead — not COGS — and manage the P&L accordingly until the ~50 agency inflection point.

§07 — Gross Margin Projection

7.1 Variable Gross Margin (Infrastructure + AI Only)

Excludes compliance overhead and fixed team costs — pure unit economics of delivering the product.

Agencies	Monthly Revenue (Solo $199 avg)	Variable COGS (×$7.41)	Variable GM $	Variable GM %
10	$1,990	$74	$1,916	**96%**
50	$9,950	$371	$9,579	**96%**
100	$19,900	$741	$19,159	**96%**
500	$99,500	$3,705	$95,795	**96%**

Variable gross margin is ~96% and flat across all scales. The product's core delivery (AI + infrastructure + SMS) costs pennies to serve once built. This is the template-based architecture dividend from CRIT-02 resolution.

7.2 Blended Gross Margin (Including Fixed COGS Allocation)

Fixed COGS includes the operations portion of engineering (~30% of $20K midpoint = $6,000/month). Compliance is excluded (operating expense, not COGS). This is the standard SaaS gross margin presentation.

Agencies	Monthly Revenue	Variable COGS	Fixed COGS (engineering ops)	Total COGS	Blended GM %
10	$1,990	$74	$6,000	$6,074	**-205%**
25	$4,975	$185	$6,000	$6,185	**-24%**
50	$9,950	$371	$6,000	$6,371	**36%**
75	$14,925	$556	$6,000	$6,556	**56%**
100	**$19,900**	**$741**	**$6,000**	**$6,741**	**66% ✓**
150	$29,850	$1,112	$6,000	$7,112	**76%**
200	$39,800	$1,482	$6,000	$7,482	**81%**
500	$99,500	$3,705	$6,500 (team growth)	$10,205	**90%**

Revenue uses $199/month (conservative Solo-tier baseline). Blended tiers ($330/month average across Solo/Team/Agency) improve all margins by ~40%.

SaaS 60% Gross Margin Threshold Crossed: between 75 and 100 agencies.

Exact Scalability Threshold (60% GM floor)

```
At 60% target: COGS ≤ 40% of Revenue
$6,000 + ($7.41 × n) = 0.40 × ($199 × n)
$6,000 = $79.60n − $7.41n = $72.19n
n = 83 agencies
```

Scalability Threshold: 83 agencies (Solo-tier baseline). 60 agencies if blended tier revenue ($330/month average).

7.3 Operating Break-Even (Full P&L)

Total monthly burn from Artifact 12 §10: $22,000–$38,000. Midpoint: $30,000/month.

Revenue per Agency	Break-Even Agency Count	ARR at Break-Even
$199 (Solo only)	$30,000 / $199 = **151 agencies**	$360K
$250 (early blended)	$30,000 / $250 = **120 agencies**	$360K
$330 (mature blended)	$30,000 / $330 = **91 agencies**	$360K

Operating break-even range: 91–151 agencies. At peer-referral GTM velocity (estimated 2–3 new agencies/month post-E1 validation), break-even is achievable within 18–24 months of first paying customer.

Total Addressable Beachhead Revenue at Break-Even

- 100–150 agencies × $330 average ARR = **$396K–$594K AUD/year**
- This is 7–11% beachhead penetration (500 eligible agencies per VI-3 estimate)
- Entirely achievable without chain expansion or EMR integration dependency

§08 — Token Efficiency Strategy (Ranked by ROI)

Strategy 1: Prompt Caching — Implement Day 1

Actions it applies to	All Haiku calls (vacancy orchestration, SPP completeness)
Cost reduction	72% on system prompt overhead (largest single token spend)
Implementation effort	**Low** — Anthropic prompt caching is a single API parameter change (`cache_control` on static system prompt blocks)
Estimated margin improvement	~$0.094/agency/month savings (from $0.127 → $0.033 on cached portion)
Recommended timeline	Sprint 1 — before first agency onboards

```
# Haiku call with prompt caching enabled
{
  "model": "claude-haiku-4-5-20251001",
  "system": [
    {
      "type": "text",
      "text": "[STATIC APP COMPLIANCE + PERSONA + SAFETY LEVELS]",
      "cache_control": {"type": "ephemeral"} // Cache this block
    },
    {
      "type": "text",
      "text": "[DYNAMIC SPP CONTEXT — DO NOT CACHE]"
    }
  ]
}
```

Note: Even at full price without caching, inference at $0.127/agency/month is not the cost problem. Caching is implemented for architectural correctness and future-proofing as agency count scales, not for current margin emergency.

Strategy 2: SMS Gateway Volume Negotiation — Priority at 50+ Agencies

Actions it applies to	All notification dispatches (3 SMS per vacancy incident)
Cost reduction	MessageMedia volume pricing: 15–25% reduction at 50+ agencies; 30–40% at 200+ agencies
Current cost	$0.08/SMS domestic (pay-as-you-go)
At 50 agencies	$0.06/SMS (volume tier) — saves $0.06/incident × 600 incidents/month = $36/month
At 200 agencies	$0.05/SMS — saves $0.12/incident × 2,400 incidents/month = $288/month
Implementation effort	**Low** — negotiation, no engineering change (Channel Wrapper architecture in place)
Recommended timeline	Initiate negotiation at 30 agencies; lock volume pricing at 50 agencies
Why SMS is the priority, not inference	SMS accounts for **46% of variable COGS** ($3.42/agency/month). A 25% reduction in SMS unit cost saves more per agency than eliminating all LLM inference costs entirely.

Strategy 3: Model Routing (Reserve Sonnet for L3 Only) — Already Specified

Actions it applies to	All L1/L2 agentic actions
Cost reduction	Already implemented in v1 architecture (CLAUDE.md HS-STRAT-03). Haiku for L1/L2, Sonnet for L3 only.
Current Sonnet cost	$0.036/agency/month for 2.5 L3 escalations — minimal at this scale
If Sonnet were used for all actions	68.5 calls × $0.01425 = $0.976/agency/month — 7× more expensive
Implementation effort	Zero — already the specified architecture
Recommended timeline	Maintained as the v1 default; revisit L3 definition in `agentic-logic-spec`
L3 scope discipline is the key guard	If the definition of "L3 action" expands in `agentic-logic-spec`, Sonnet costs will increase. VACANCY_UNRESOLVED is the only current L3 action. Any new L3 classification must be justified against the cost impact.

Strategy 4: SLM Distillation for SPP Completeness Checks — v2, Post-Validation

Actions it applies to	Daily SPP completeness analysis (Action Type C — 30 calls/month/agency)
Cost at Haiku	$0.037/agency/month — not material
Cost at custom SLM	~$0.003/agency/month (92% reduction)
Implementation effort	**High** — requires training dataset generation from Haiku completeness judgements, fine-tuning pipeline, evaluation framework

Estimated margin improvement at 500 agencies: $0.034/agency × 500 = $17/month — immaterial at this scale

Recommended timeline	Year 2 only if SPP complexity increases significantly (e.g., SBIs added in v2 requiring more sophisticated completeness analysis)

Current verdict: SLM distillation has near-zero ROI at beachhead scale. The SPP completeness check costs $0.037/month/agency. This is not a bottleneck. Do not invest engineering time on SLM distillation in v1 or v2 unless agency count exceeds 1,000 and SPP schema complexity grows substantially.

Strategy 5: Async Batch Processing for SPP Analysis — Not Applicable in v1

Actions it applies to	Daily SPP completeness checks (non-time-sensitive)
Cost reduction	50% on async-eligible calls
Implementation effort	Medium
Verdict	SPP completeness is already non-time-sensitive. Batch processing would save $0.018/agency/month. Not worth implementation overhead at beachhead scale. Reconsider at 500+ agencies.

§09 — Infrastructure Scaling Roadmap

Scale	Architecture	AI Stack	Estimated Monthly Infra Cost (AUD)	Notes
0–50 agencies (Beachhead)	Serverless (Lambda + DynamoDB + S3). Single-region ap-southeast-2. Managed services only. No ops overhead.	Anthropic API (Haiku + Sonnet). Prompt caching enabled Day 1. MessageMedia SMS gateway.	**$200–$750**	Predominantly Lambda invocation + DynamoDB on-demand + SMS. Well within Artifact 12 estimate.
50–200 agencies (Early Growth)	Add read replicas for SPP query performance. S3 Intelligent Tiering for 7-year audit log. Consider DynamoDB Reserved Capacity.	Same AI stack. Add model routing classifier (lightweight Lambda). Volume SMS pricing negotiated.	**$750–$3,000**	Infra cost grows sub-linearly with agencies (shared fixed costs amortize).
200–1,000 agencies (Scale)	ElastiCache for SPP match query caching. Scheduled batch processing for nightly SPP completeness. Multi-AZ failover for compliance continuity.	Add Claude prompt cache warming for predictable daily patterns (SPP checks). Evaluate Haiku fine-tuning for agency-specific shortlist formatting.	**$3,000–$12,000**	At this scale, infra is 3–4% of revenue — healthy SaaS ratio. Compliance fixed costs have fully amortized.
1,000+ agencies (Expansion)	Evaluate multi-region (ap-southeast-2 + ap-southeast-4 Melbourne) for resilience. API gateway throttling per-agency. Advanced CloudTrail anomaly detection.	Evaluate custom SLM distillation for SPP completeness if SPP schema grows. Vector store for longitudinal SPP pattern analysis (v3 feature).	**$12,000+**	Model pricing deflation is likely by this stage. Token costs at scale remain sub-1% of revenue.

Scaling cost-per-agency trajectory

Scale	Revenue/Agency	Infra/Agency	Infra % of Revenue
10 agencies	$199	$20.00 (fixed dominates)	10.1%
50 agencies	$199	$7.41 (variable only)	3.7%
200 agencies	$250 (blended)	$4.00 (volume discounts)	1.6%
1,000 agencies	$280 (blended, mature)	$3.50	1.3%

Infrastructure becomes an increasingly negligible share of revenue as scale increases. **The margin story at 200+ agencies is excellent.**

§10 — Gross Margin Optimization Recommendations

Recommendation 1 — Raise Average Revenue Per Agency (ARPA) Before 83-Agency Threshold

Current state	Modelled at $199/month (Solo tier — conservative beachhead assumption)
Target state	$330/month average (blended Solo + Team + Agency tiers)
Mechanism	At E1 close, present the Agency Owner the "Team tier" value proposition: Angela manages 3 coordinators — 3 seats × $149 = $447/month. The moat story (SPP knowledge portable across coordinators) is the Agency Owner value prop. Every Solo-tier agency that upgrades to Team-tier cuts the break-even agency count.
Impact	Raises threshold from 83 agencies (Solo-only) to ~50 agencies (blended). **33 fewer agencies needed to cross 60% GM.**
Implementation owner	PM Lead (pricing conversation at E1 close)
Timeline	Before first commercial subscription (Q2 2026)

Recommendation 2 — Negotiate Volume SMS Pricing at 30-Agency Milestone

Current state	$0.08/SMS pay-as-you-go (MessageMedia / AWS SNS)
Target state	$0.055–$0.06/SMS at volume tier (30–50 agencies committed)
Mechanism	SMS is 46% of variable COGS. A 25–30% reduction moves agency-level COGS from $7.41 → $6.33. At 100 agencies, this saves $108/month in margin — more than eliminating all LLM inference costs.
AWS SNS vs. MessageMedia	AWS SNS (ap-southeast-2) pricing is $0.00581 USD per 1,000 SMS for Australia (~$0.009 AUD per SMS at current rates) — significantly cheaper than MessageMedia. Evaluate SNS as primary gateway; MessageMedia as failover.
Impact	~$1.08/agency/month variable COGS reduction; ~1.5 percentage points GM improvement at 100 agencies.
Implementation owner	Engineer (gateway switch) + PM Lead (contract)
Timeline	Evaluate at 20 agencies; switch by 30 agencies

Recommendation 3 — Charge for SPP Onboarding Session as a Positive Margin Event

Current state	Concierge SPP migration is founder time (unpriced at beachhead)
Target state	$500–$1,500 one-time SPP migration session fee (Artifact 12 §11 secondary revenue stream)
Mechanism	The onboarding session is not a cost centre — it is the product's cold-start mitigation and the moat seeding event. Charging for it:

- Recovers founder time cost
- Signals premium positioning (care quality costs money)

- Filters agencies who are not serious about the SPP (low-value churners)
- Partially offsets the compliance amortization cost during the sub-25-agency phase

Impact	\$500–\$1,500 per new agency = \$1,000 average × 12 new agencies in Year 1 = \$12,000 AUD in Year 1 onboarding revenue — covers ~0.4 months of compliance overhead
Implementation owner	PM Lead (pricing framing at XP-2A LOI meetings)
Timeline	Q2 2026 — frame in the first LOI conversation

§11 — Partnership Economics Integration (PAC → CAC → LTV)

Carrying PAC estimates from Artifact 18 §7 into the unit economics model.

11.1 Customer Acquisition Cost (CAC) Model

Acquisition Channel	CAC Components	Estimated CAC (AUD)	Notes
Concierge (E1/E2 — Angela, Tom)	Founder time only	$0 cash (time cost)	Pre-revenue — CAC is opportunity cost, not cash
Peer referral (GTM-2)	No direct spend	$0–$500 (referral dinner/event)	GTM assumption to validate at E1 close
ACCPA Advisory (post-beachhead)	Advisory board equity + annual fee	$2K cash + 0.25% equity / ÷ agencies enabled	If ACCPA endorsement unlocks 20 agencies: $100/agency + equity
ACCPA Conference / Innovation Awards	Sponsorship + PM time	$10K per event / ÷ agencies acquired	If conference yields 5 agencies: $2,000/agency (high)
ACCPA Commercial Endorsement	$30K–$60K/year	$300–$600/agency (at 100 agencies/year)	Target Year 2+ only; validates after beachhead
AlayaCare integration (if VI-1 confirmed)	$30–60K engineering equivalent + ongoing	$300–$600 amortized per agency (at 100 agencies over 2 years)	High cost; only if AlayaCare is the distribution path

11.2 Customer Lifetime Value (LTV) Model

Assumes 24-month agency average lifetime (conservative — SPP data gravity should extend this substantially)

Tier	Monthly ARR	Onboarding Fee	24-Month LTV	LTV/CAC at Peer Referral
Solo	$199/month	$750 avg	$199 × 24 + $750 = **$5,526**	$5,526 / $250 = **22:1**
Team (avg 2.5 seats × $149)	$373/month	$1,000 avg	$373 × 24 + $1,000 = **$9,952**	$9,952 / $250 = **40:1**
Agency (avg 7 seats × $119)	$833/month	$1,500 avg	$833 × 24 + $1,500 = **$21,492**	$21,492 / $500 = **43:1**

LTV/CAC ratios are excellent at peer referral CAC. The economics compress significantly if AlayaCare integration is required (CAC rises to $300–$600, LTV/CAC narrows to 9–18:1 — still acceptable for SaaS but not exceptional).

11.3 SPP Data Gravity LTV Extension Effect

The SPP moat changes the LTV model materially. Per Artifact 19 §6:

- By Month 6: switching cost is real (agency loses months of structured preference history)
- By Month 18: switching cost is structural (years of familiarity data, match quality is demonstrably better)

Conservative assumption (24-month lifetime) is likely understated. If SPP data gravity holds (XP-3B probe validates > 4-week rebuild cost), a more realistic churn model is:

- Month 0–6: Normal SaaS churn risk (product trust building)
- Month 6–18: Churn rate drops to <5%/year (data gravity engaging)
- Month 18+: Churn approaches structural minimum (switching cost outweighs any competitor feature gap)

At 36-month average lifetime (conservative for agencies with 18+ months of SPP data):

- Solo LTV: $199 × 36 + $750 = **$8,214**
- LTV/CAC at peer referral: **33:1**

The SPP moat is not just a positioning claim — it is a LTV multiplier. Every month of SPP data accumulated is a retention asset that extends the revenue relationship.

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§12 — HS-STRAT-03 Handshake: Token Budget → Model Selection

Per CLAUDE.md Article V: this handshake transfers the Max Token Budget per Agentic Action to `agentic-logic-spec` *for model selection specification.*

ID	Agentic Action	Autonomy Level	Model Specified	Token Budget (Input + Output)	Cost per Action	Rationale
HS-STRAT-03a	L1/L2 Vacancy orchestration (state management, HITL coordination)	L1 Informer / L2 Verifier	**Claude Haiku 4.5 (`claude-haiku-4-5-20251001`)**	2,200 tokens (1,550 system + 450 dynamic + 200 output)	$0.0015	Routine, repetitive, narrow input/output structure. Budget comfortably supports Haiku at any realistic scale.
HS-STRAT-03b	L2 SPP completeness analysis	L1 Informer	**Claude Haiku 4.5**	2,050 tokens (1,550 + 350 + 150)	$0.00123	Structured analysis, daily frequency, well within Haiku capability.
HS-STRAT-03c	L3 Vacancy unresolved escalation (VACANCY_UNRESOLVED decision)	L3 Escalator	**Claude Sonnet 4.6 (`claude-sonnet-4-6`)**	3,500 tokens (1,550 + 1,150 + 800)	$0.01650	Complex, low-frequency, high-stakes (coordinator must understand why matching failed). Sonnet reasoning quality justified.
HS-STRAT-03d	Carer briefing assembly (structured SPP field assembly, P-3 branch)	L1 Informer	**No LLM — template only**	0 tokens	$0.00	CRIT-02 resolution confirmed. Template-based in v1.
HS-STRAT-03e	Notification content (SMS to carer, client, family)	L1 Informer	**No LLM — template only**	0 tokens	$0.00	CRIT-02 resolution confirmed. String interpolation only.
HS-STRAT-03f	Audit log entry	L1 Informer	**No LLM**	0 tokens	$0.00	Structured write. APPAuditLogEntry schema is deterministic.

`agentic-logic-spec` model selection rule (binding from this artifact)

```
IF action.level IN ['L1', 'L2'] AND action.type NOT IN ['VACANCY_UNRESOLVED']:
    USE claude-haiku-4-5-20251001
    ASSERT token_budget <= 2200
    ASSERT cost_per_action <= $0.002 AUD

IF action.level == 'L3' OR action.type == 'VACANCY_UNRESOLVED':
    USE claude-sonnet-4-6
    ASSERT token_budget <= 3500
    ASSERT cost_per_action <= $0.017 AUD

IF action.type IN ['NOTIFICATION_DISPATCH', 'BRIEFING_ASSEMBLY', 'AUDIT_LOG']:
    USE NO_LLM
    ASSERT content_generation_method == 'TEMPLATE_INTERPOLATION'
    // Enforcement: CRIT-02 compliance guard — NFR-INJ-* requirements
```

Reject criteria for `agentic-logic-spec`: Any logic spec that:

- Assigns Sonnet to L1/L2 actions (cost violation — would add $0.013/action premium for no quality gain)
- Assigns Haiku to L3 VACANCY_UNRESOLVED (quality risk — coordinator needs Sonnet reasoning quality for complex escalation context)
- Uses LLM for notification content or briefing assembly (CRIT-02 violation — CRITICAL compliance finding)

§13 — v2 Cost Inflection Points

When the following v2 features unlock (per Artifact 17 §07 V2 Unlock Register), the cost model changes:

Feature	Unlock Condition	Cost Impact	New Monthly COGS Component
LLM-generated external notifications	NFR-INJ-01–04 implemented + red-team tested	Adds LLM inference to every notification	+$0.58/agency/month at Haiku (3 notifications × 12 incidents × ~600 tokens each) — immaterial
WhatsApp channel (CRIT-01 v2 unlock)	APP 8 legal review complete	WhatsApp Business API pricing (Twilio: ~$0.04–$0.06/message) vs. SMS ($0.05–$0.08)	Roughly cost-neutral or marginally cheaper; architectural improvement, not cost driver
Carer App (in-app push)	APP 8 confirmation + engineering build	Replaces SMS ($0.05–$0.08/message) with push ($0.00005/push via SNS) — 1,000× cheaper	**Major cost reduction: SMS drops from $3.42 → $0.001/agency/month.** At 500 agencies, this saves $1,710/month. The Carer App is not just a UX upgrade — it is a **5–7 percentage point gross margin improvement** at scale.
AlayaCare bi-directional integration (AX-02 confirmed)	Write-back endpoint confirmed	Additional API calls to AlayaCare per assignment (~$0.001/call)	Negligible — AlayaCare API calls are a trivial cost
SPP v2 (LLM-assisted field extraction from coordinator notes)	DPIA v2 + CRIT-04 resolved	Haiku call for extracting structured SPP fields from coordinator input	+$0.003/agency/month — immaterial
Structured Baseline Indicators (SBI — Future Discovery)	Separate discovery cycle + DPIA	New PHI fields; compliance infrastructure expansion required	Compliance fixed cost increases by ~$1,500–$2,000/month (additional DPIA + ongoing counsel for clinical monitoring category)

Key v2 finding: The Carer App is not just a product feature — it is the biggest single gross margin lever available. Replacing SMS ($3.42/agency/month) with in-app push ($0.03/agency/month) reduces variable COGS by 99% and saves 46% of current per-agency delivery cost. At 500 agencies, this is worth $16,950/month in recovered margin — far more than any AI inference optimisation.

§14 — Financial Summary

Single-Page Executive View

Metric	Value	Source
Variable COGS per agency/month (v1)	**$7.41 AUD**	§05
AI inference component	**$0.13 (1.7% of COGS)**	§02
SMS / channel component	**$3.42 (46% of COGS)**	§04
Infrastructure component	**$3.86 (52% of COGS)**	§03
Variable gross margin	**96%**	§07.1
Blended GM at 100 agencies	**66% (above 60% floor)**	§07.2
Scalability Threshold (60% GM)	**~83 agencies (Solo) / ~60 agencies (blended)**	§07.2
Operating break-even	**91–151 agencies**	§07.3
LTV/CAC (peer referral channel)	**22:1–43:1 by tier**	§11.2
Dominant cost driver	**SMS, not AI**	§05
Largest v2 margin lever	**Carer App (replace SMS)**	§13
Token budget per L1/L2 action	**2,200 tokens / $0.0015**	§12
Token budget per L3 action	**3,500 tokens / $0.0165**	§12
Year 1 PAC (Tier 1 partnerships, cash)	**$20–90K AUD**	Artifact 18 §7

§15 — Handshake Table — HS-STRAT-03 (Unit Economics → Execution)

ID	From (This Artifact)	To (Execution Plugin)	What Must Transfer
HS-STRAT-03a	§12 — Haiku budget: 2,200 tokens, L1/L2	`agentic-logic-spec` model selection	Haiku is the mandatory model for all L1/L2 actions. Any spec that uses Sonnet for L1/L2 is a cost violation. `harness-audit-grader` should flag Sonnet assignments to L1/L2 actions as a cost architecture defect.
HS-STRAT-03b	§12 — Sonnet budget: 3,500 tokens, L3 only	`agentic-logic-spec` model selection	Sonnet is reserved exclusively for L3 VACANCY_UNRESOLVED. Adding new L3 actions requires PM Lead sign-off and a cost impact assessment against this artifact.
HS-STRAT-03c	§12 — No LLM for notifications/briefings	`agentic-logic-spec` NFR	Template enforcement is a CRIT-02 compliance constraint, not just a cost choice. The logic spec must assert `content_generation_method == TEMPLATE_INTERPOLATION` for all external-facing content.
HS-STRAT-03d	§13 — Carer App is the v2 margin lever	`create-prd` v2 roadmap	Carer App is not optional infrastructure — it is the single largest gross margin improvement available in v2. The PRD v2 section should note the $16,950/month margin recovery at 500 agencies. This is a financial priority, not just a UX upgrade.
HS-STRAT-03e	§14 — Scalability Threshold: 83 agencies	`create-prd` business context	PRD §1 business context: product becomes unit-economics positive (>60% GM) at 83 agencies. This is the commercial milestone that defines Series A readiness.

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§16 — Action Items

Due	Owner	Action	Priority
Sprint 1	Engineer	Implement Anthropic prompt caching on system prompt static blocks for all Haiku calls. Cache key = `[app_compliance_block + persona_block + safety_levels_block]`. Dynamic SPP context MUST NOT be cached.	**HIGH**
Sprint 1	Engineer	Confirm AWS SNS (ap-southeast-2) SMS pricing vs. MessageMedia. If SNS is cheaper at current volume, implement SNS as primary gateway (MessageMedia as failover). Channel Wrapper architecture already supports this switch.	**HIGH**
2026-04-01	PM Lead	Frame onboarding (SPP migration) session as paid service ($750–$1,500) in all XP-2A LOI conversations. Present as "knowledge architecture session" — the event that starts the moat.	**HIGH**
2026-04-01	PM Lead	Present Agency tier pricing rationale to agency owners with ≥ 2 coordinators. Every Solo → Team upgrade reduces the break-even count by ~0.7 agencies. At Angela's agency (3 coordinators × $149 = $447/month), ARR is 2.2× a Solo agency.	**HIGH**
2026-05-01	PM Lead + Engineer	Validate Artifact 20 COGS model against actuals from first 5 agencies. Specifically confirm: (1) average incidents/month per agency; (2) SMS volume vs. projected 36/month; (3) L3 escalation frequency vs. projected 2.5/month. Re-run §07 with actual data.	**MEDIUM**
2026-06-01	PM Lead	At 20+ agencies: initiate MessageMedia / AWS SNS volume pricing conversation. Target $0.055–$0.06/SMS. 25% reduction in SMS cost yields 0.8 percentage points GM improvement.	**MEDIUM**
Year 2	Engineer + PM Lead	Begin Carer App infrastructure planning. At 100+ agencies, replacing SMS with in-app push recovers $342/month in variable COGS and grows to $16,950/month at 500 agencies. This is the highest-ROI v2 engineering investment available.	**HIGH (Year 2)**

Artifact integrity note: This model is grounded in confirmed v1 architecture decisions — rule-based matching (Artifact 12), template-based notifications (Artifact 17 CRIT-02), AU-hosted SMS gateway (Artifact 17 CRIT-01), and partnership PAC estimates (Artifact 18). The model should be re-run after first 5 paying agencies to replace the four key assumptions confirmed here: (1) 12 vacancy incidents/month/agency, (2) 2.5 L3 escalations/month/agency, (3) 3 SMS recipients per incident, and (4) 80% prompt cache hit rate.

Per CLAUDE.md Article IV Rule 6: this is the final Strategy Plugin skill. The Strategy Plugin pipeline is now complete. The Execution Plugin (pm-execution) begins with `create-prd`.

Strategy Plugin — Pipeline Completion Status

```
■ Startup Canvas (12) ← context contract
■ SWOT Analysis (13) ← BUILD+DEFEND signal
■ Value Proposition (14) ← "What After" → OKR
■ User Journey Map (15) ← Moment of Truth + L3 interventions
■ Compliance Privacy Audit (16) ← 4 CRITs + 8 HIGHs
■ Remediation Decision Record (17) ← Stage 4 gate cleared
■ Partnership Mapping (18) ← Moat Summary → Positioning
■ Positioning Statement (19) ← Geoffrey Moore + ERRC + Moat Assessment
■ AI Unit Economics (20) ← Token budget → model selection ← THIS ARTIFACT

■ NEXT: Execution Plugin → create-prd (Artifact 21)
         Inputs: HS-STRAT-01 (Moment of Truth + L3 interventions from Artifact 15)
                 HS-STRAT-02 (Compliance NFRs from Artifact 16)
                 HS-STRAT-04 ("What After" OKRs from Artifact 14)
                 HS-STRAT-05 (Positioning from Artifact 19)
                 HS-STRAT-03 (Token budget + model selection from this artifact)
```

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